

1. Introduction to Optimization

Optimization T2026

Master in Fundamental Principles of Data Science



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Introduction

- **What is Optimization?** Given a system or process, find the **best solution** possibly subject to **constraints**.
- **Objective Function:** Indicator of “goodness” of the solution of the optimization problem, e.g., cost, profit, time, etc.
- **Decision Variables:** Variables that influence process behavior and can be adjusted for optimization.
- We are interested in a **systematic approach** to the optimization process, and to make it as efficient as possible.

Current applications

How do we make the **best use of limited resources**?

- **Artificial intelligence:** train neural networks and language models by **minimizing a loss function** or **maximizing expected reward**.
- **Logistics:** choose delivery routes that **minimize travel time** while respecting capacity and delivery windows.
- **Energy:** schedule electricity generation/battery storage to **meet demand at minimum cost**.
- **Chip design:** place components to **reduce wiring and power use** while meeting performance requirements.
- **Finance:** choose investments that balance **expected return, risk, transaction costs**.
- ... and many more

The common structure: **decision variables + objective + constraints.**

First introductory examples

Problem. *Find the solution of*

$$\text{minimize } f(x, y) = (e^x - 1)^2 + (y - 1)^2.$$

This is an example of an **unconstrained** optimization problem.

The set where we must look for the solution, the **feasible set**, is the entire two-dimensional space $\mathbb{R}^2$.

In this problem, the **objective function** is f and the **decision variables** are x, y .

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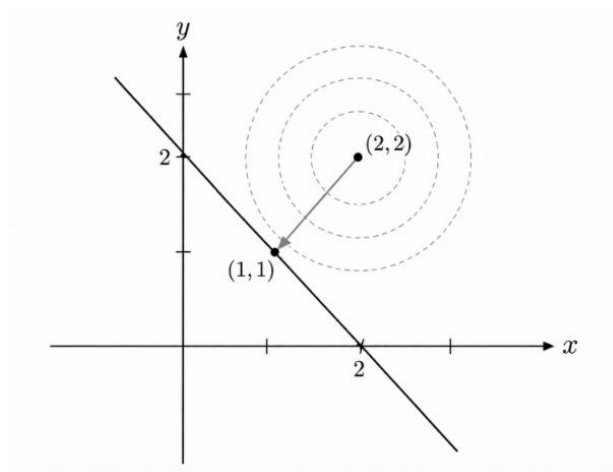
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The solution (**optimizer**) is $(x^*, y^*)^T = (0, 1)^T$: *the function value is zero only at this point and is positive elsewhere.*

First introductory examples

Problem. Find the point on the line $x + y = 2$ that is closest to the point $(2, 2)^T$.



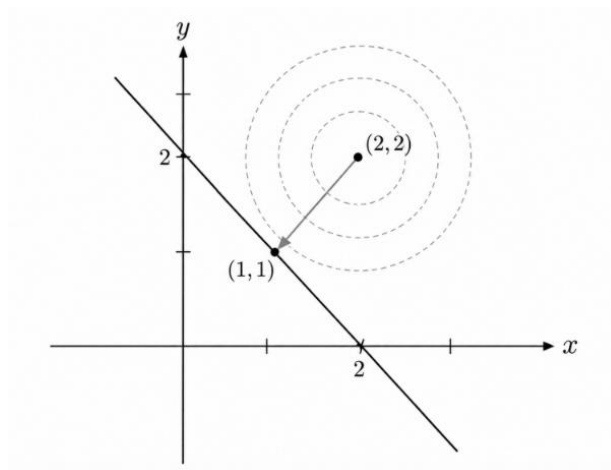
The **mathematical model** can be written as

$$\begin{array}{ll} \text{minimize} & f(x, y) = (x - 2)^2 + (y - 2)^2, \\ \text{subject to} & x + y = 2. \end{array}$$

In this example, the **objective function** is f , the **decision variables** are x , y , and the **(feasible)** set is defined by an equality.

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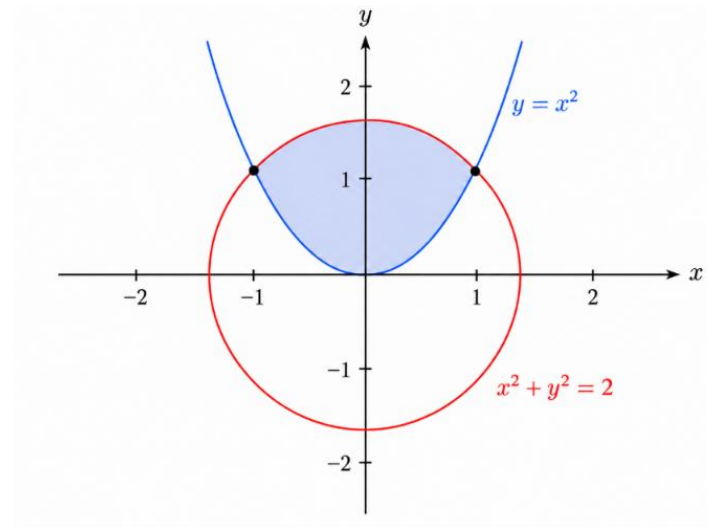
The solution is $(x^*, y^*)^T = (1, 1)^T$.

First introductory examples

Problem. Find the point such that:

$$\begin{array}{ll}\text{minimize} & f(x, y) = x, \\ \text{subject to} & x^2 \leq y, \\ & x^2 + y^2 \leq 2.\end{array}$$

In this example, the (feasible) set where we must look for the solution is given by two constraints defined by inequalities.

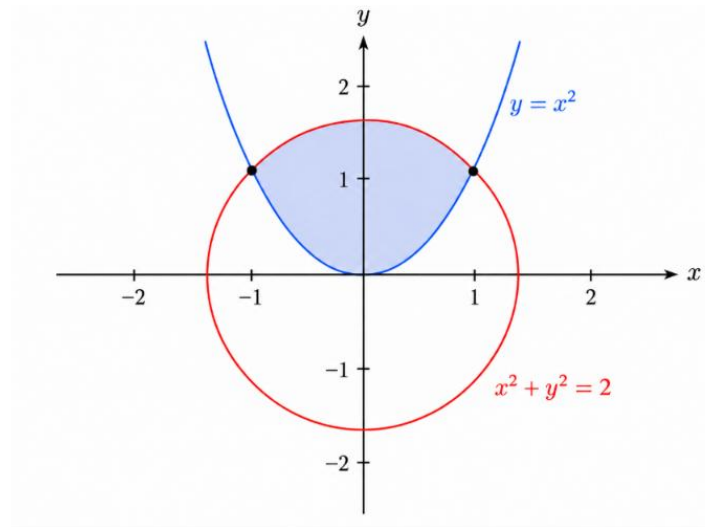


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The solution is $(x^*, y^*)^T = (-1, 1)^T$.

Optimization viewpoints

- **Mathematical viewpoint:** What can we guarantee?
Existence and uniqueness of minimizers; optimality conditions; convergence and convergence rates.
- **Computational viewpoint:** How do we compute a solution?
Algorithm design and implementation; numerical stability; time and memory costs.
- **Modelling viewpoint:** Are we solving the right problem?
Objectives and constraints; data uncertainty; sensitivity and validation of results.

Current research: large-scale and nonconvex optimization, optimization for machine learning, decisions under uncertainty, ...

In this course: fundamental concepts and methods.

General setup

Definition:

The **general nonlinear optimization** (NLO) problem can be written as follows:

$$\begin{array}{ll}\min & f(x), \\ \text{subject to} & g_i(x) = 0, \quad i \in I = \{1, \dots, m\}, \\ & h_j(x) \leq 0, \quad j \in J = \{1, \dots, p\}, \\ & x \in \mathcal{C},\end{array}$$

where $x \in \mathbb{R}^n$, $\mathcal{C} \subset \mathbb{R}^n$ is a certain set, and $f, g_1, \dots, g_m, h_1, \dots, h_p$ are real-valued functions defined on $\mathcal{C}$.

Terminology:

- The function f is called the **objective function** of the NLO.
- The set $\mathcal{F}$ defined by:

$$\mathcal{F} = \{x \in \mathcal{C} : g_i(x) = 0, i = 1, \dots, m, h_j(x) \leq 0, j = 1, \dots, p\},$$

is called the **feasible set** (or **feasible region**).

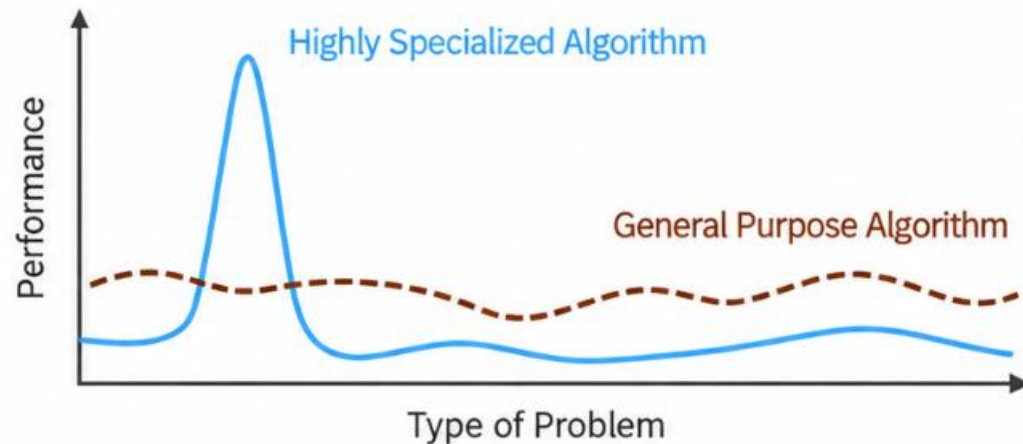
- If $\mathcal{F} = \emptyset$ then we say that the optimization problem is **infeasible**.
- If the infimum of f over $\mathcal{F}$ is attained at $x^* \in \mathcal{F}$, then we call x^* an **optimal solution** of the NLO, and $f(x^*)$ the **the optimal (objective) value of the NLO**.

There is no universally best algorithm!

$$\begin{cases} \text{minimize}_{\mathbf{x}} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{x} \in \mathcal{F} \end{cases}$$

No Free Lunch Theorem (Wolpert, Macready, 1997):

Given a finite set $\mathcal{F}$ and a finite set S of real numbers, assume that $f : \mathcal{F} \rightarrow S$ is chosen at random according to uniform distribution on the set $S^{\mathcal{F}}$ of all possible functions from $\mathcal{F}$ to S . For the problem of optimizing f over the set $\mathcal{F}$, no algorithm performs better than blind search (i.e., pick at random and check).



⇒ Useful optimization requires some assumptions on f , such as continuity, convexity, differentiability, or other problem structure.

Classification of optimization problems

- **Unconstrained Optimization:** The index sets I and J are empty:

$$g_1 = \dots = g_m = h_1 = \dots = h_p = 0,$$

and $\mathcal{C} = \mathbb{R}^n$.

- **Linear Optimization (LO)** (Linear programming): The functions $f, g_1, \dots, g_m, h_1, \dots, h_p$ are linear (affine: $F(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$) and the set $\mathcal{C}$ either equals to $\mathbb{R}^n$, the positive (negative) orthant $\mathbb{R}_+^n$, or is polyhedral.

- **Quadratic Optimization (QO):** The objective function f is quadratic:

$$f(\mathbf{x}) = \mathbf{x}^T Q \mathbf{x} + \mathbf{c}^T \mathbf{x} + d,$$

all the constraint functions $g_1, \dots, g_m, h_1, \dots, h_p$ are linear and the set $\mathcal{C}$ is $\mathbb{R}^n$ or the positive (negative) orthant $\mathbb{R}_+^n$, Q is a $n \times n$ real matrix ($Q \in \mathbb{R}^{n \times n}$), $\mathbf{c} \in \mathbb{R}^n$, and $d \in \mathbb{R}$.

- **Quadratically Constrained Quadratic Optimization:** Same as QO, except that the constraint functions are quadratic.
- **Convex Quadratic Optimization (CQO).**
- **Convex Quadratically Constrained Quadratic Optimization:**
- ...

The infinite-dimensional optimization problem

There is a more **general nonlinear optimization infinite-dimensional** problem that can be written as follows:

Find the state function $x(t)$ and the control function $u(t)$ such that

$$\begin{array}{ll} \text{minimize} & \mathcal{J} = \Phi(t_0, x_0, t_f, x_f) + \int_{t_0}^{t_f} F(t, x(t), u(t)) dt, \\ \text{subject to} & t \in [t_0, t_f], \\ & \dot{x}(t) = f(t, x(t), u(t)), & \text{(dynamic constraints),} \\ & b_L \leq b(t_0, x_0, t_f, x_f) \leq b_U, & \text{(boundary conditions),} \\ & g_L \leq g(t, x(t), u(t)) \leq g_U, & \text{(path constraints),} \\ & u_L \leq u(t) \leq u_U, & \text{(control constraints).} \end{array}$$

In the **multi-objective** optimization problem, the cost function $\mathcal{J}$ may involve several independent quantities, this is:

$$\vec{\mathcal{J}} = (\mathcal{J}_1 \cdots \mathcal{J}_M)^T.$$

The infinite-dimensional optimization problem

There are multiple solution approaches for the infinite-dimensional optimization problem. They are commonly divided into:

- **Indirect methods.** The initial problem is transformed into a Hamiltonian boundary-value problem that must be solved. These methods use the calculus of variations (Pontryagin's Maximum Principle).
- **Direct methods.** The original problem is first discretized and then re-written as a finite-dimensional nonlinear optimization problem (NLO).